

To summarize, therefore, the approach to network analysis should be to determine whether component voltages and currents are known, in which case a direct approach to the analysis may be made using the principles observed by Kirchhoff's laws. If this is not possible then network reduction should be tried in order that the network is sufficiently simplified that it becomes manageable. Should all else fail, then Kirchhoff's laws should be applied to derive loop simultaneous equations from which the solution will be obtained.

4.3

Mesh analysis

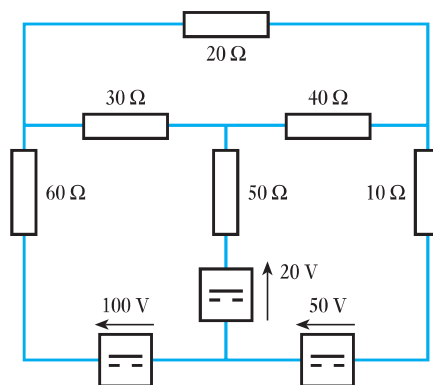
This method is given a number of different names – all of which are an indication of the analysis technique employed. It is variously known as Maxwell's circulating current method, loop analysis or Mesh current analysis. The terminology is chosen to distinguish it from the familiar 'branch current' technique, in which currents are assigned to individual branches of a circuit. The branch current method was first introduced in Chapter 3 and has been used hitherto. Mesh analysis, of course, relies on Kirchhoff's laws just the same. The technique proceeds as follows:

- Circulating currents are allocated to closed loops or meshes in the circuit rather than to branches.
- An equation for each loop of the circuit is then obtained by equating the algebraic sum of the e.m.f.s round that loop to the algebraic sum of the potential differences (in the direction of the loop, mesh or circulating current), as required by Kirchhoff's voltage (second) law.
- Branch currents are found thereafter by taking the algebraic sum of the loop currents common to individual branches.

Example 4.10

Calculate the current in each branch of the network shown in Fig. 4.13.

Fig. 4.13 Circuit diagram for Example 4.10



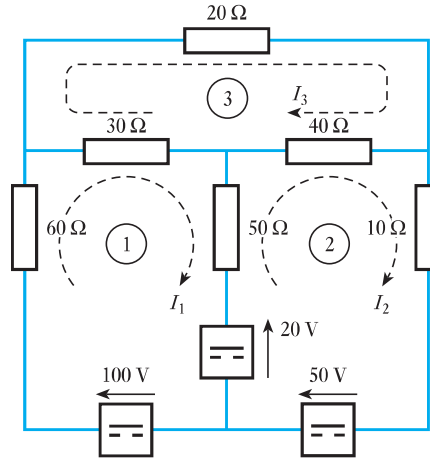
Let the circulating loop currents be as shown in Fig. 4.14.

In loop ①:

$$100 - 20 = I_1(60 + 30 + 50) - I_2 50 - I_3 30$$

$$\therefore 80 = 140I_1 - 50I_2 - 30I_3 \quad (a)$$

Fig. 4.14 Circuit diagram for Example 4.10



In loop ②:

$$50 + 20 = I_2(50 + 40 + 10) - I_1 50 - I_3 40$$

$$\therefore 70 = -50I_1 + 100I_2 - 40I_3 \quad (b)$$

In loop ③:

$$0 = I_3(30 + 20 + 40) - I_1 30 - I_2 40$$

$$\therefore 0 = -30I_1 - 40I_2 + 90I_3 \quad (c)$$

Solving for these equations gives

$$I_1 = 1.65 \text{ A} \quad I_2 = 2.16 \text{ A} \quad I_3 = 1.50 \text{ A}$$

Current in $60 \Omega = I_1 = 1.65 \text{ A}$ in direction of I_1

Current in $30 \Omega = I_1 - I_3 = 0.15 \text{ A}$ in direction of I_1

Current in $50 \Omega = I_2 - I_1 = 0.51 \text{ A}$ in direction of I_2

Current in $40 \Omega = I_2 - I_3 = 0.66 \text{ A}$ in direction of I_2

Current in $10 \Omega = I_2 = 2.16 \text{ A}$ in direction of I_2

Current in $20 \Omega = I_3 = 1.50 \text{ A}$ in direction of I_3 .

In Example 4.10 all the circulating loop currents have been taken in the same direction (i.e. clockwise). This is not essential when using this method, but if the same direction is adopted for the loop currents then the equations will always be of the form:

$$E_1 = R_{11}I_1 - R_{12}I_2 - R_{13}I_3 \dots - R_{1n}I_n$$

$$E_2 = -R_{21}I_1 + R_{22}I_2 - R_{23}I_3 \dots - R_{2n}I_n$$

$$E_3 = -R_{31}I_1 - R_{32}I_2 + R_{33}I_3 \dots - R_{3n}I_n$$

$$E_n = -R_{n1}I_1 - R_{n2}I_2 - R_{n3}I_3 \dots + R_{nn}I_n$$

where E_1 = the algebraic sum of the e.m.f.s in loop ① in the direction of I_1 ;

E_2 = the algebraic sum of the e.m.f.s in loop ② in the direction of I_2 ,

etc.;

R_{11} = sum of resistances in loop ①;

R_{22} = sum of resistances in loop ②, etc.;

R_{12} = total resistance common to loops ① and ②;

R_{23} = total resistance common to loops ② and ③, etc.

By their definitions $R_{12} = R_{21}$, $R_{23} = R_{32}$, etc. Note that in the equation derived from each loop it is only the term in the loop's own circulating current that is positive.

By observing these rules the equations necessary for the solution of the circuit problem can be written down by inspection of the circuit. This can be confirmed by examination of equations (a), (b) and (c) in Example 4.10.

4.4

Nodal analysis

This technique of circuit solution, also known as the Node Voltage method, is based on the application of Kirchhoff's first (current) law at each junction (node) of the circuit, to find the node voltages. It should be noted that, in contrast, both the branch current and Mesh current techniques of circuit analysis are based on the applications of Kirchhoff's second (voltage) law, often to find unknown currents.

The Node Voltage method generally proceeds as follows:

Step 1: Choose a reference node to which all node voltages can be referred.

Label all the other nodes with (unknown) values of voltage, V_1 , V_2 , etc.

Step 2: Assign currents in each connection to each node, except the reference node, in terms of the node voltages, V_1 , V_2 , etc.

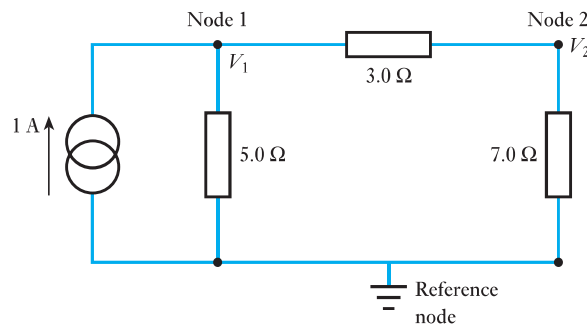
Step 3: Apply Kirchhoff's current law at each node, obtaining as many equations as there are unknown node voltages.

Step 4: Solve the resulting equations to find the node voltages.

Example 4.11

Using Nodal analysis, calculate the voltages V_1 and V_2 in the circuit of Fig. 4.15.

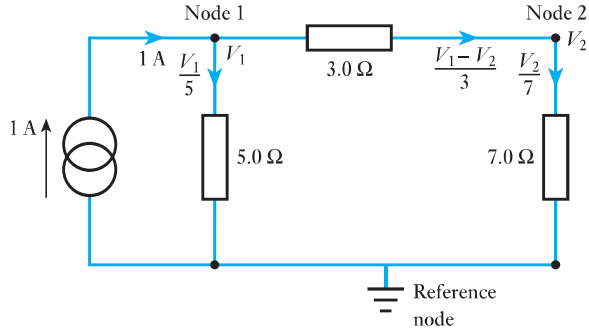
Fig. 4.15 Network for Example 4.11



Refer to the four steps previously indicated:

Step 1: Reference node chosen. Voltages V_1 and V_2 assigned to the other two nodes.

Step 2: Assign currents in each connection to each node (Fig. 4.16).

Fig. 4.16 Part of Example 4.11

Step 3: Apply Kirchhoff's current law to sum the currents at each node.

At node 1:

$$\frac{V_1}{5} + \left(\frac{V_1 - V_2}{3} \right) = 1$$

which can be simplified to

$$V_1 \left(\frac{1}{5} + \frac{1}{3} \right) - \frac{V_2}{3} = 1 \quad (a)$$

At node 2:

$$\frac{V_1 - V_2}{3} = \frac{V_2}{7}$$

which simplifies to

$$\frac{V_1}{3} - V_2 \left(\frac{1}{3} + \frac{1}{7} \right) = 0 \quad (b)$$

Step 4: Solve node voltage equations (a) and (b).

From equation (b), by multiplying each term by 21,

$$7V_1 - V_2(7 + 3) = 0$$

$$\therefore 7V_1 = 10V_2$$

$$\text{so } V_2 = \frac{7}{10}V_1 \quad (c)$$

From equation (a), by multiplying each term by 15,

$$8V_1 - 5V_2 = 15 \quad (d)$$

Substitute for V_2 , from equation (c), in equation (d):

$$8V_1 - \frac{35V_1}{10} = 15$$

$$\text{so } 4.5V_1 = 15$$

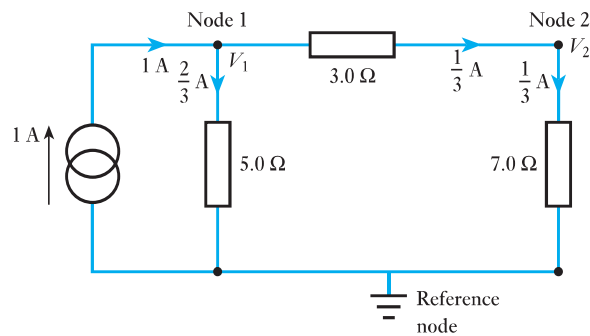
$$V_1 = \frac{10}{3} \text{ V}$$

From (c)

$$V_2 = \frac{7}{3} \text{ V}$$

To check the accuracy of the calculation, see for yourself if Kirchhoff's current law is obeyed for each node. It will be seen that the currents are as in the circuit of Fig. 4.17.

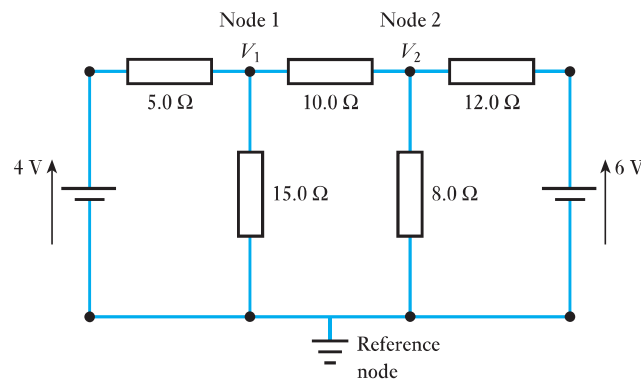
Fig. 4.17 Part of Example 4.11



Example 4.12

Using the Node Voltage method calculate the voltages V_1 and V_2 in Fig. 4.18 and hence calculate the currents in the 8Ω resistor.

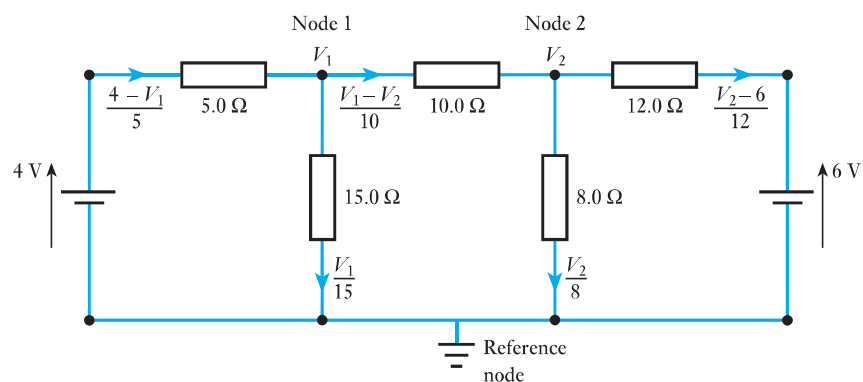
Fig. 4.18 Network for Example 4.12



Step 1: Reference node shown. Voltages V_1 and V_2 assigned.

Step 2: Assign currents in each connection to each node (Fig. 4.19).

Fig. 4.19 Part of Example 4.12



Step 3: Apply Kirchhoff's current law at each node.

At node 1:

$$\frac{4 - V_1}{5} = \frac{V_1 - V_2}{10} + \frac{V_1}{15}$$

Multiply each term by 30:

$$24 - 6V_1 = 3V_1 - 3V_2 + 2V_1$$

$$11V_1 - 3V_2 = 24 \quad (a)$$

At node 2:

$$\frac{V_1 - V_2}{10} = \frac{V_2 - 6}{12} + \frac{V_2}{8}$$

Multiply each term by 120:

$$12V_1 - 12V_2 = 10V_2 - 60 + 15V_2$$

$$12V_1 - 37V_2 = -60 \quad (b)$$

Step 4: Solve for V_1 and V_2 .

Equation (a) $\times \frac{12}{11}$ gives:

$$12V_1 - \frac{36V_2}{11} = \frac{24 \times 12}{11} \quad (c)$$

Equation (c) – equation (b) gives

$$33.37V_2 = 86.18$$

$$V_2 = 2.55 \text{ V}$$

From (a)

$$11V_1 = 24 + 3 + 2.55 = 31.65$$

$$V_1 = 2.88 \text{ V}$$

Hence the current in the 8Ω resistor is

$$\frac{V_2}{8} = 0.32 \text{ A}$$

A second method of solving this problem by Nodal analysis, using source conversion techniques, is shown in section 4.7.

4.5

Superposition theorem

The Superposition theorem states that in any network containing more than one source, the current in, or the p.d. across, any branch can be found by considering each source separately and adding their effects: omitted sources of e.m.f. are replaced by resistances equal to their internal resistances.

This sounds very complicated, but is really quite simple when demonstrated by example. Example 4.13 illustrates the manner in which Example 4.7 would be solved by means of the Superposition theorem.